Hospital Volume and Outcomes after Elective Abdominal Aortic Aneurysm Repair: A Systematic Review and Dose–Response Meta-Analysis

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Background: Procedural volume has been associated with outcomes after abdominal aortic aneurysm (AAA) repair, yet the magnitude and shape of the volume–outcome relationship remain uncertain. This systematic review and dose–response meta-analysis evaluated the association between hospital and surgeon procedural volume and outcomes following elective AAA repair.

Methods: MEDLINE, Embase, and the Cochrane Library were systematically searched from inception to February 9, 2026. Studies evaluating the association between procedural volume and outcomes after elective AAA repair were included. Dose–response meta-analyses were performed using restricted cubic spline models to assess potential linear and nonlinear associations between procedural volume and outcomes.

Results: Eleven observational studies were analyzed. Higher hospital procedural volume was significantly associated with lower overall mortality, with a predominantly linear relationship ($n = 202,212$; $\beta = -0.0102$, $P = 0.011$). Hospitals at the 90th percentile of procedural volume had 35% lower odds of mortality compared with those at the 10th percentile (odds ratio: 0.65; 95% confidence interval: 0.44–0.96). A significant inverse association was also observed between hospital volume and failure-to-rescue after open repair ($n = 60,096$; $\beta = -0.0407$, $P = 0.0006$), although the P90 versus P10 contrast was not statistically significant due to wide confidence intervals.

Conclusion: Higher hospital procedural volume is associated with lower mortality after elective AAA repair and reduced failure-to-rescue after open repair. These findings suggest that institutional experience plays an important role in outcomes after AAA repair, supporting strategies aimed at centralizing care in higher-volume centers.

Registration: PROSPERO CRD4202612850863.

Ethical approval: Ethical approval was not required for this study, as it is a systematic review and meta-analysis based exclusively on previously published data and did not involve the collection of individual patient data.

Funding: This study received no financial or nonfinancial support.

Conflict of interest: The authors declare no conflicts of interest.

Data availability: No datasets were generated during the conduct of this review. Additional data and materials related to the analysis are available upon reasonable request.

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Ann Vasc Surg 2026; 131: 368–382
https://doi.org/10.1016/j.avsg.2026.06.025
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Manuscript received: April 25, 2026; manuscript accepted: June 17, 2026; published online: 25 June 2026

INTRODUCTION

The relationship between procedural volume and surgical outcomes has been one of the most consistently investigated organizational determinants of quality in complex surgery. Across multiple specialties, higher hospital and surgeon volumes have been associated with lower perioperative mortality, better management of complications, and improved overall postoperative outcomes, likely reflecting greater technical experience, multidisciplinary support, and institutional readiness to deliver complex care.[1–4]

In vascular surgery, this volume–outcome relationship has been particularly relevant in abdominal aortic aneurysm (AAA) repair, a procedure that demands careful patient selection, advanced operative expertise, and the ability to promptly recognize and manage perioperative complications. Both open surgical repair (OSR) and endovascular aneurysm repair (EVAR) have become established treatment strategies for elective AAA, but their outcomes may be influenced not only by patient- and anatomy-related factors but also by the experience of the treating hospital and surgeon.[5–7] This has supported increasing interest in centralization policies and the concentration of AAA care in high-volume centers.[5,6]

Societies have incorporated this concept into clinical guidance. The European Society for Vascular Surgery (ESVS), for example, recommends that elective AAA repair be performed in centers with sufficient annual procedural volume and appropriate expertise, regardless of whether an open or endovascular approach is used.[8] However, the evidence supporting specific volume thresholds remains heterogeneous. Prior observational studies have used different definitions of low- and high-volume practice, variably examined hospital versus surgeon volume, and inconsistently stratified results according to repair technique. In addition, mortality has been reported using different endpoints, including in-hospital and 30-day mortality, which further complicates cross-study interpretation.[1,9]

As a result, although a volume–outcome association in AAA repair is widely acknowledged, the magnitude and consistency of this relationship remain incompletely defined, particularly when comparing hospital volume versus surgeon volume and EVAR versus open repair. In this context, this systematic review and meta-analysis aimed to evaluate early mortality and failure-to-rescue (FTR) outcomes according to hospital and surgeon procedural volume in elective AAA repair.

METHODS

Protocol and Registration

We performed the systematic review and meta-analysis according to the Cochrane Handbook for Systematic Reviews of Interventions and structured it according to the Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines.[10,11] The study protocol was registered in the International Prospective Register of Systematic Reviews (PROSPERO) under registration number CRD420261285086.[12]

Eligibility Criteria

In this dose–response meta-analysis, studies were eligible if they met all of the following criteria: (1) enrolled adult patients undergoing elective AAA repair; (2) compared procedural volumes using at least 3 quantitative exposure categories to enable dose–response modeling; (3) reported mortality or FTR; and (4) provided sufficient quantitative data to estimate study-specific dose–response slopes using the method of Greenland and Longnecker.[13]

Studies were excluded if they (1) included ruptured AAA repairs; (2) did not stratify outcomes according to hospital or surgeon procedural volume; (3) had no full text available; or (4) were conference abstracts, case reports, trial registrations without available results, reviews, meta-analyses, or animal studies.

Search Strategy and Study Selection

We systematically searched MEDLINE, Embase, and Cochrane Library databases from inception to February 9th, 2026. Our PICO framework was: P: patients undergoing elective AAA repair; I: low-volume hospital; C: high-volume hospital; O: mortality and failure to rescue. The search strategy for all searched databases was (''abdominal aortic aneurysm'' OR ''abdominal aortic aneurysm'' OR ''abdominal aortic'' OR ''aortic aneurysm'' OR AAA) AND (''elective'' OR ''elective surgery'') AND (''open repair'' OR ''open repairs'' OR ''open surgical repair'' OR ''open abdominal aortic aneurysm repair'' OR ''open abdominal aortic surgery'' OR ''open aortic aneurysm repair'' OR ''endovascular aneurysm repair'' OR EVAR) AND (''hospital volume'' OR ''surgeon volume'' OR ''provider volume'' OR ''case volume'' OR ''center volume'' OR ''volume-outcome'' OR ''volume outcome'' OR ''high volume'' OR ''low volume'' OR volume). We also searched the references of the included

studies and previous systematic reviews and meta-analyses aiming for the inclusion of additional studies.[14]

Two authors (M.L.C and L.M.D.) independently conducted the search, imported results into Rayyan, a web-based systematic review tool, and triaged the studies. After the exclusion of duplicates and titles/abstracts unrelated to the clinical question, the eligibility of each remaining study was assessed based on the review of the full-text articles. Disagreements were resolved by consensus.

Data Extraction

Two authors (M.L.C. and L.M.D.) independently extracted data from the included studies using a standardized data extraction form. The following information was collected: (1) general study characteristics, including first author, year of publication, study design, country, study period; (2) patient and sample characteristics, including number of patients and follow-up duration; and (3) reported our outcome of interest, as defined a priori in the study protocol.

For the dose–response meta-analysis using the method of Greenland and Longnecker, it was necessary to extract the following data from each study: dose midpoint, effect estimate, lower and upper confidence intervals (CIs), log odds ratio (OR), standard error, and the number of cases and participants in each exposure category. Studies were excluded if these data were not explicitly reported or if insufficient information was provided to calculate or infer them. In addition, at least 3 quantitative exposure categories of procedural volume were required to allow proper estimation of the dose–response relationship.

Outcome of Interest

The outcomes of interest were mortality (defined as 30-day or in-hospital mortality) and FTR, defined as death following a postoperative complication, representing the inability to prevent mortality after an adverse event.

Volume Definition Across Included Studies

Substantial heterogeneity was observed across studies in the definition of procedural volume thresholds and in the categorization of exposure levels, including the use of tertiles, quartiles, quintiles, and sextiles. To improve comparability and enable quantitative modeling, a prespecified harmonization approach was applied. Whenever possible, procedural volume was standardized to procedures per year, with annualization performed when cumulative volumes were reported over a defined study period. Analyses were conducted separately for hospital and surgeon volume, with additional stratification by repair type (OSR and EVAR). For categorical exposure data, the midpoint of each category was assigned as the exposure dose. For open-ended categories, the midpoint was estimated by assuming a category width equivalent to that of the adjacent category.

Statistical Analysis

A 2-stage random-effects dose–response meta-analysis was conducted using the Greenland and Longnecker method.[13] This method allows estimation of a study-specific linear trend from summarized data reported across exposure categories, while accounting for the fact that the category-specific risk estimates within each study are correlated because they usually share a common reference group. This approach allows evaluation of how the risk of an outcome changes across increasing levels of procedural volume, rather than relying only on comparisons between arbitrarily defined low- and high-volume groups. In the first stage, study-specific log ORs (log ORs) and corresponding covariance matrices were reconstructed from the reported categorical exposure data. These covariance matrices account for the correlation between risk estimates across exposure categories within the same study, since all category-specific comparisons are typically made against a shared reference group and are therefore not statistically independent. Their reconstruction is required to appropriately estimate study-specific dose–response slopes and standard errors. Reconstruction required category-specific information on exposure levels and on the distribution of participants and events across categories. When category sizes were not explicitly reported, they were derived from the published data whenever possible, including totals combined with reported percentages or category-specific event and nonevent counts. If category sizes were not reported and could not be reliably derived, covariance matrices were not reconstructed, and the study was not included in the corresponding dose–response analysis. In the second stage, pooled estimates were obtained using restricted maximum likelihood. Potential nonlinearity in the association between hospital volume and outcomes was evaluated using restricted cubic splines with prespecified knots. Nonlinearity was assessed by testing the spline coefficients against a linear model. Results

were expressed as ORs with 95% CIs, using the 10th percentile of the volume distribution as the reference. Clinically meaningful contrasts (e.g., 90th vs. 10th percentile) were also reported. Between-study heterogeneity was quantified using Cochran's Q test, τ^2, and the I^2 statistic.

Additional Analysis

To investigate potential sources of heterogeneity, mixed-effects meta-regression analyses were performed using study-specific dose–response slopes (log OR per unit increase in procedural volume) as the dependent variable. Prespecified study-level covariates included study period, publication year, repair type (open surgical repair [OSR] versus EVAR), proportion of male patients, and the reported incidence of cardiac, respiratory, and renal complications. The proportion of between-study variance explained by each moderator was quantified using R^2, following standard recommendations for meta-regression analyses.[15] A two-sided P value < 0.05 was considered statistically significant.

The volume associated with the lowest predicted risk was estimated from the fitted restricted cubic spline model as a model-derived point of minimum predicted risk, with substantial uncertainty, rather than as a definitive or ''optimal'' clinical threshold. Statistical uncertainty around this estimate was evaluated using parametric simulation based on the variance–covariance matrix of the model coefficients.[16]

Leave-one-out approach, in which each study was sequentially excluded, was also used to assess the influence of individual studies on the pooled estimates.

Risk of Bias Assessment

Two independent reviewers (M.L.C. and L.M.D.) assessed the Risk of Bias in Non Randomized Studies of Interventions (ROBINS-I V2) tool.[17] Disagreements were resolved through consensus. Publication bias could not be assessed adequately because the power of this test is insufficient to discriminate between chance and true funnel plot asymmetry when analyzing fewer than 10 studies.[18]

RESULTS

Study Selection

As detailed in Supplementary Figure 1, the initial search identified 741 records. After removal of 202 duplicate records and screening of titles and abstracts, 93 studies were selected for full-text review according to the prespecified eligibility criteria. Of these, 82 studies were excluded for the following reasons: conference abstracts ($n = 29$), failure to compare high- and low-volume centers ($n = 21$), insufficient data for dose–response analysis ($n = 19$), lack of outcomes of interest ($n = 5$), mixed or inappropriate populations ($n = 3$), duplicate reports ($n = 2$), evaluation of a different exposure ($n = 1$), review article ($n = 1$), and other methodological issues ($n = 1$). Supplementary Material Ilists the studies excluded at the full-text screening stage and the corresponding reasons for exclusion.

Study Characteristics

A total of 11 studies, published between 2009 and 2025, were included, comprising 329,701 patients, of whom 157,772 underwent EVAR and 171,929 underwent OSR.[3,4,19–27] Follow-up ranged from in-hospital assessment to 365 days. Detailed study characteristics are presented in Table I.

Hospital Volume Definitions

Table II shows substantial heterogeneity in how procedural volume was defined across studies, including variation in hospital versus surgeon volume, repair-specific versus overall AAA volume, number of strata, and cut-off values. This variability highlights the limitations of direct threshold comparison across studies and supports the use of a dose–response approach.

Pooled Analyses

Hospital-volume analysis

Overall mortality. In the dose–response meta-analysis including 6 studies ($n = 202{,}212$), a statistically significant association between hospital volume and overall mortality was observed ($\chi^2 = 6.40$, df = 2, $P = 0.041$; $I^2 = 90.9\%$). The linear spline component indicated a significant inverse relationship ($\beta = -0.0102$, $P = 0.011$), suggesting lower odds of mortality with increasing procedural volume. The nonlinear spline component was not statistically significant ($P = 0.324$), indicating no evidence of departure from linearity (Fig. 1).

Hospitals at the 90th percentile of procedural volume had 35% lower odds of mortality than those at the 10th percentile (OR 0.65; 95% CI 0.44–0.96; Supplementary Fig. 2). Although the spline model estimated the lowest predicted mortality at approximately 59 procedures per year, this value was located near the upper boundary of the observed data and should therefore be interpreted cautiously.

Table I. Study and patients characteristics across included studies

First author, year	Study design	Country	Period	No. of patients		Male, %		Mean or median age, years		Length of follow-up, days
				EVAR	Open	EVAR	Open	EVAR	Open	
D'Oria, 2021	Multinational observational cohort study	International	2010–2016	37,630	11,174	81.8	81.8	72.9	72.9	In hospital
Dubois, 2018	Retrospective cohort	Canada	2005–2014	0	7,211	NA	79.2	NA	71	365
Eckstein, 2007	Post-hoc analysis of a prospective surgeon-led national registry	Germany	1999–2004	0	10,163	NA	NA	NA	67.5	In hospital
Gray, 2020	Administrative cohort	UK	2011–2019	22,962	8,867	87.9	88.8	75.9	71.1	30–90[a]
Holt, 2009	Administrative cohort	UK	2005–2007	1,645	5,668	85.3	84.2	73.9	72.2	In-hospital[b]
McPhee, 2011	Administrative cohort	USA	2003–2007	8,121	5,972	82.1	74.9	75	72	In-hospital[b]
Meuli, 2025	Registry cohort	Switzerland	2019–2022	128	333	81.2	83.8	75	71.2	365
Nimptsch, 2017	Administrative cohort	Germany	2009–2014	41,678	22,227	NA	NA	NA	NA	In-hospital[b]
Rosero, 2017	Retrospective cohort	USA	2001–2011	0	47,223	NA	67.1	NA	NA	In-hospital
Scali, 2022	Retrospective analysis of prospectively collected Vascular Quality Initiative (VQI) registry data	USA	2003–2020	0	14,068	NA	74	NA	69.2	In-hospital
Trenner, 2018	Nationwide retrospective secondary data analysis	Germany	2005–2013	45,608	39,023	NA	NA	NA	NA	In-hospital

AAA, abdominal aortic aneurysm; EVAR, endovascular aneurysm repair; NA, not available; UK, United Kingdom; USA, United States of America.
[a]Follow-up ranged from 30-90 days.
[b]Outcomes were assessed during the index hospitalization (in-hospital follow-up).

Table II. Definitions of procedural volume strata across included studies

First author, year	Strata definition	Definition of volume	Volume subgroups	Patients in each group
D'Oria, 2021	Hospital annual volume quartiles	Procedures per year	EVAR: Q1 1−22.5 Q2 22.5−37.7 Q3 37.7−63.7 Q4 63.7−121 open: Q1 1−11 Q2 11−23 Q3 23−43 Q4 43−116	EVAR: Q1 369 Q2 416 Q3 367 Q4 476 open Q1 740 Q2 640 Q3 548 Q4 144
Dubois, 2018	Surgeon volume quintiles[c]	Procedures per year	Very low: median 3 Low: median 13 Medium: median 22 High: median 30 Very high: median 54	• Very low: 475 • Low: 777 • Medium: 1,252 • High: 1,781 • Very high: 2,926
Eckstein, 2007	Hospital annual volume sextiles	Procedures per year	S1: 1−9 S2: 10−19 S3: 20−29 S4: 30−39 S5: 40−49 S6: ≥50	S1: 367 S2: 1,881 S3: 1,634 S4: 1,551 S5: 739 S6: 3,991
Gray, 2020	Surgeon and hospital annual volume quintiles	Procedures per year	S1 ≤ 20, S2 = 20−39, S3 = 40−59, S4 = 60−79, S5 = 80−99, S6 > 100	S1 = 1,009 S2 = 5,309 S3 = 5,874 S4 = 10,213 S5 = 3,696 S6 = 5,728
Holt, 2009	Hospital AAA volume quintiles	Procedures per year	Overall: Q1 < 21, Q2 = 22−27, Q3 = 27.5−42, Q4 = 42.5−73.5, Q5 = 74−129 EVAR Q1 < 8, Q2 = 9−15, Q3 = 16−26.5, Q4 = 27−65.5, Q5 = 66−115 OSR Q1 < 15.5, Q2 = 16−23, Q3 = 24−32, Q4 = 33−43, Q5 = 43.5−76.5	Q1 = 1,431 Q2 = 1,470 Q3 = 1,554 Q4 = 1,287 Q5 = 1,571
McPhee, 2011	Surgeon and hospital volume quintiles	Procedures per year	EVAR: Low ≤ 15, medium = 16−70, high > 70 Open: Low < 7, medium = 7−30, high > 30	Low = 3,064 Medium = 8,149 High = 2,860
Meuli, 2025	Hospital annual volume tertiles[d]	Procedures per year	Low < 5, medium 6−9, high ≥ 10	Low = 162 Medium = 87 High = 212
Nimptsch, 2017	Hospital AAA volume sextiles	Procedures per year	EVAR: Q1 = 6 (IQR 3−9), Q2 = 17 (IQR 15−19), Q3 = 26 (IQR 24−30), Q4 = 40 (36−45), Q5 = 64 (57−75) Open: Q1 = 3 (IQR 1−4), Q2 = 9 (IQR 7−10), Q3 = 15 (13−17), Q4 = 21 (19−25), Q5 = 39 (33−46)[a]	Very Low = 12,703 Low = 12,762 Medium = 12,718 High = 12,729 Very High = 12,992

(*Continued*)

Table II. Continued

First author, year	Strata definition	Definition of volume	Volume subgroups	Patients in each group
Rosero, 2017	Hospital annual volume quartiles	Procedures per year	2001[b] - Q1 0−4 Q2 > 4−11 Q3 > 11 − 25 Q4 > 25 2011 Q1 0−2 Q2 > 2−6 Q3 > 6−13 Q4 > 13	Q1 1,681 Q2 4,630 Q3 10,370 Q4 30,552
Scali, 2022	Hospital annual volume quartiles[e]	Procedures per year	Q1: ≤3 Q2: 4−6 Q3: 7−10 Q4: >10	Q1: 409 Q2: 958 Q3: 1,956 Q4: 6,468
Trenner, 2018	Hospital annual volume quartiles	Procedures per year	Q1 2−5 Q2 8−13 Q3 19−27 Q4 42−82	Q1 2,258 Q2 8,663 Q3 20,719 Q4 52,991

AAA, abdominal aortic aneurysm; EVAR, endovascular aneurysm repair; FEVAR, fenestrated endovascular aneurysm repair; IQR, interquartile range; OSR, open surgical repair.
[a]Values represent median and IQR of annual procedures for each study quintile.
[b]For the abbreviated period May−September 2001, centers were classified as low-volume if they performed <4 total procedures during that interval.
[c]Data stratified by hospital volume were not available.
[d]Hospital volume was defined using annual case-volume tertiles (F/B-EVAR and/or OAR combined). Centers performing <5 cases/year were classified as low-volume, 5 to <10 cases/year as medium-volume, and ≥10 cases/year as high-volume.
[e]Hospital volume was defined according to the Society for Vascular Surgery (SVS) guidelines. Centers performing fewer than 10 open aortic procedures per year were classified as low-volume, whereas centers performing ≥10 open aortic procedures per year were classified as high-volume.

This apparent minimum may partly reflect truncation of the exposure range, since the model had limited information beyond that point to determine the shape of the association. Accordingly, the findings are more consistent with a progressive reduction in mortality with increasing hospital volume than with the identification of a discrete clinical threshold (Supplementary Fig. 7).

Open REPAIR Mortality. In the dose−response meta-analysis including 6 studies ($n = 52{,}696$), no significant association between hospital volume and mortality after open repair was observed ($\chi^2 = 3.49$, df = 2, $P = 0.175$; $I^2 = 88.5\%$). Neither the linear nor the nonlinear spline components were statistically significant. Although the likelihood ratio test suggested improved fit of the spline model compared with the linear model, the Wald test for nonlinearity was nonsignificant ($P = 0.976$), indicating no evidence of departure from linearity (Fig. 2A).

Hospitals at the 90th percentile of procedural volume had similar odds of mortality compared with those at the 10th percentile (OR 0.88; 95% CI 0.42−1.86; Supplementary Fig. 3). The spline model suggested a minimal predicted mortality at approximately 49 procedures per year (Supplementary Fig. 8). However, the CI included the null value and the curve remained relatively flat across the observed volume range, indicating no clear dose−response relationship between hospital volume and mortality for open repair.

EVAR mortality. In the dose−response meta-analysis including 5 studies ($n = 71{,}534$), no significant association between hospital volume and mortality after EVAR was observed ($\chi^2 = 2.26$, df = 2, $P = 0.323$; $I^2 = 74.6\%$). Neither the linear nor the nonlinear spline components were statistically significant. Comparison of the restricted cubic spline model with the linear model showed no significant improvement in model fit (likelihood ratio test $P = 0.136$), and the Wald test for nonlinearity was also nonsignificant ($P = 0.842$), indicating no evidence of departure from linearity (Fig. 2B).

Hospitals at the 90th percentile of procedural volume had similar odds of mortality compared with those at the 10th percentile (OR 0.89; 95% CI 0.48−1.67; Supplementary Fig. 4). The spline model suggested the lowest predicted mortality at approximately 57.5 procedures per year (Supplementary Fig. 9). However, the CI crossed the null value and the curve remained relatively flat across the observed range of procedural volumes, indicating no clear dose−response relationship.

Failure-to-rescue. In the dose−response meta-analysis including 3 studies ($n = 60{,}096$), a statistically significant association between hospital volume and FTR after open repair was observed

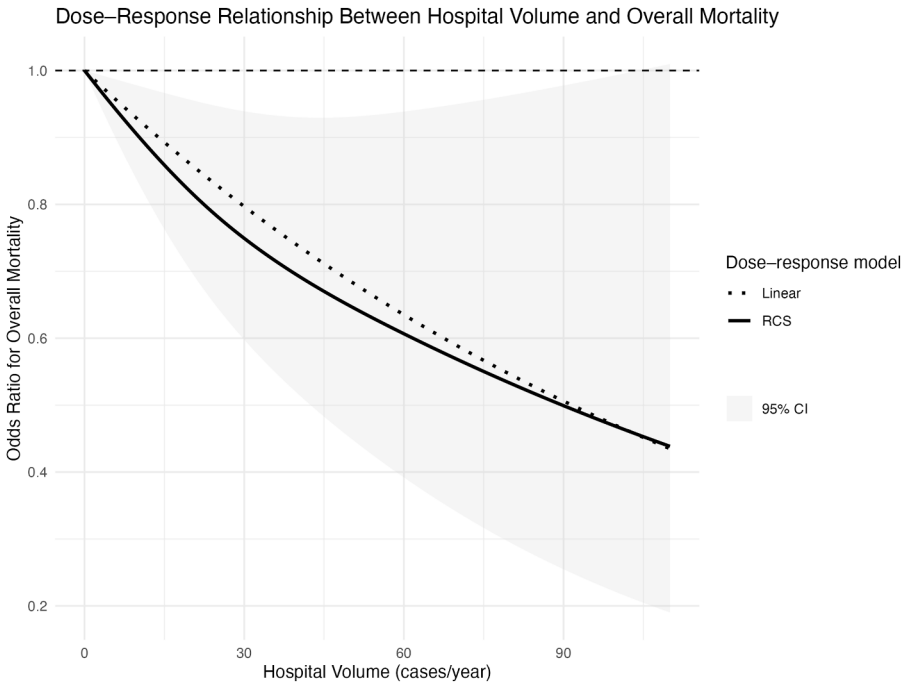


Fig. 1. Dose–response relationship between hospital volume and overall mortality estimated using restricted cubic splines. The *dotted line* represents the linear trend and the *shaded area* the 95% CI. CI, confidence interval.

($\chi^2 = 14.94$, df = 2, $P = 0.0006$; $I^2 = 84.2\%$). The primary spline term indicated a significant inverse relationship ($\beta = -0.0407$, $P = 0.0006$), suggesting lower odds of FTR with increasing hospital volume. No evidence of nonlinearity was detected (likelihood ratio test $P = 0.974$; Wald test $P = 0.567$), indicating a predominantly linear association (Fig. 3).

Hospitals at the 90th percentile of procedural volume had lower odds of FTR than those at the 10th percentile (OR 0.07; 95% CI 0.0005–9.96; Supplementary Fig. 5); however, this contrast was not statistically significant, as the CI crossed the null value. The spline model suggested the lowest predicted risk at approximately 32.6 procedures per year (Supplementary Fig. 10).

Surgeon-Volume Analysis

Open mortality. In the dose–response meta-analysis including 3 studies (n = 24,021), no statistically significant association between surgeon procedural volume and mortality after open repair was observed. The spline model suggested a decreasing trend in mortality with increasing surgeon volume, although the association did not reach statistical significance (Fig. 4).

When comparing surgeons at the 90th versus the 10th percentile of procedural volume, the pooled OR was 0.65 (95% CI 0.23–1.82), suggesting a potential reduction in mortality at higher surgeon volumes; however, the CI crossed the null value, indicating that the association was not statistically significant (Supplementary Fig. 6).

The spline model suggested the lowest predicted mortality at approximately 21.5 procedures per year (Supplementary Fig. 11). However, the wide CIs indicate substantial statistical uncertainty, and these findings should therefore be interpreted with caution.

Meta-regression. We observed a significant positive linear correlation between mean year of data collection and EVAR mortality ($P < 0.001$). For the remaining analysis, none showed a statistically significant correlation. Additional exploratory meta-regression analyses were performed for FTR using study-level covariates (Supplementary Figs. 16–19), evaluating the proportion of male patients and the rates of cardiac, respiratory, and renal complications as potential modifiers of the FTR dose–response relationship. None of these covariates were significantly associated with the dose–response slope. All meta-regression analyses are detailed in Supplementary Figures 12–19.

Leave-One-Out Analysis

Leave-one-out sensitivity analyses demonstrated that the pooled estimates remained stable after

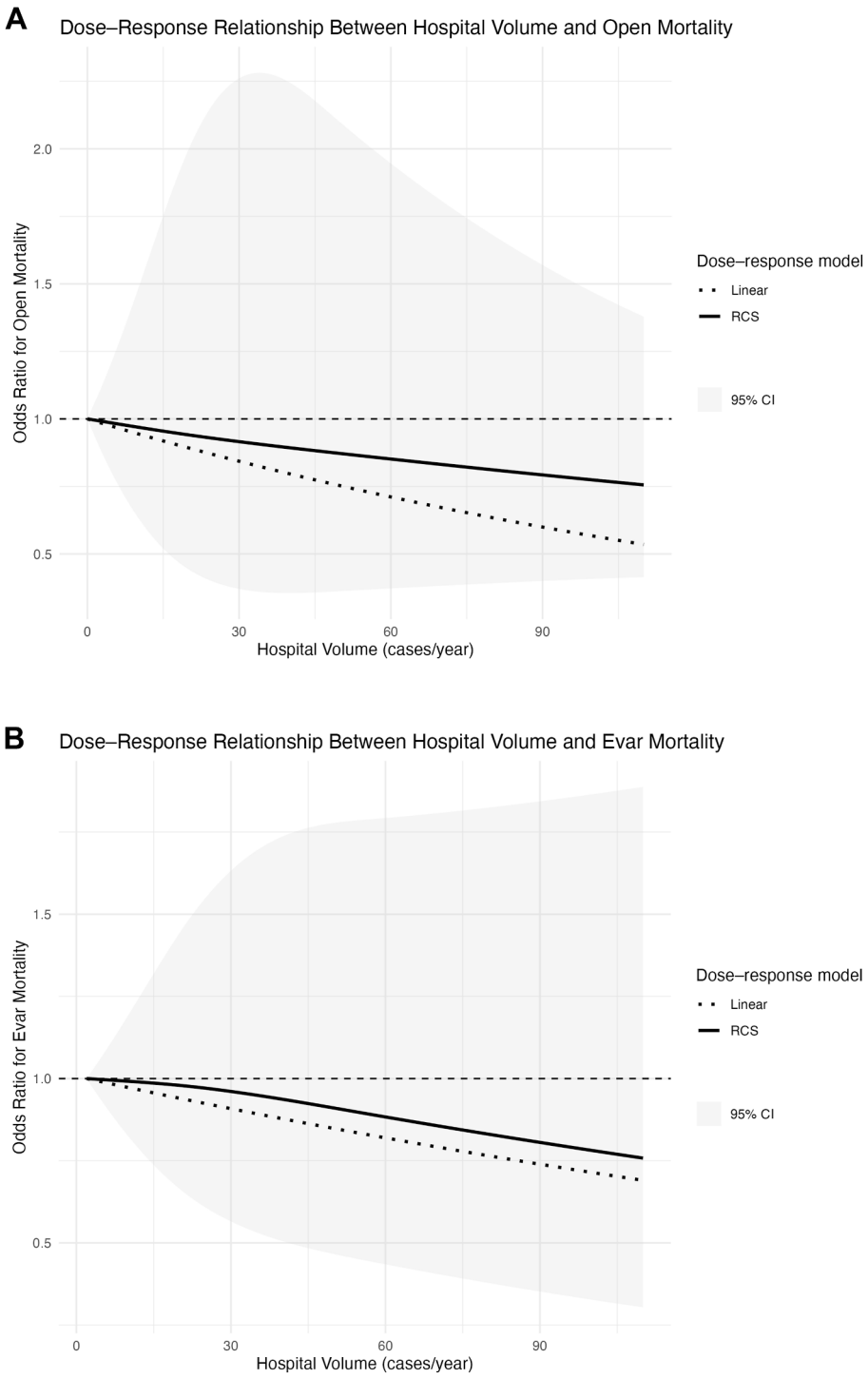


Fig. 2. Dose–response relationship between hospital volume and **(A)** open repair; and **(B)** EVAR mortality estimated using restricted cubic splines. The *dotted line* represents the linear trend and the *shaded area* the 95% CI. CI, confidence interval; EVAR, endovascular aneurysm repair.

sequential exclusion of individual studies. For overall mortality, effect estimates ranged from OR 0.91 to 0.96 per +10 procedures, while for open repair mortality estimates ranged from approximately OR 0.87 to 0.93 per +10 procedures. These findings indicate that no single study disproportionately influenced the observed associations between hospital procedural volume and mortality (Supplementary

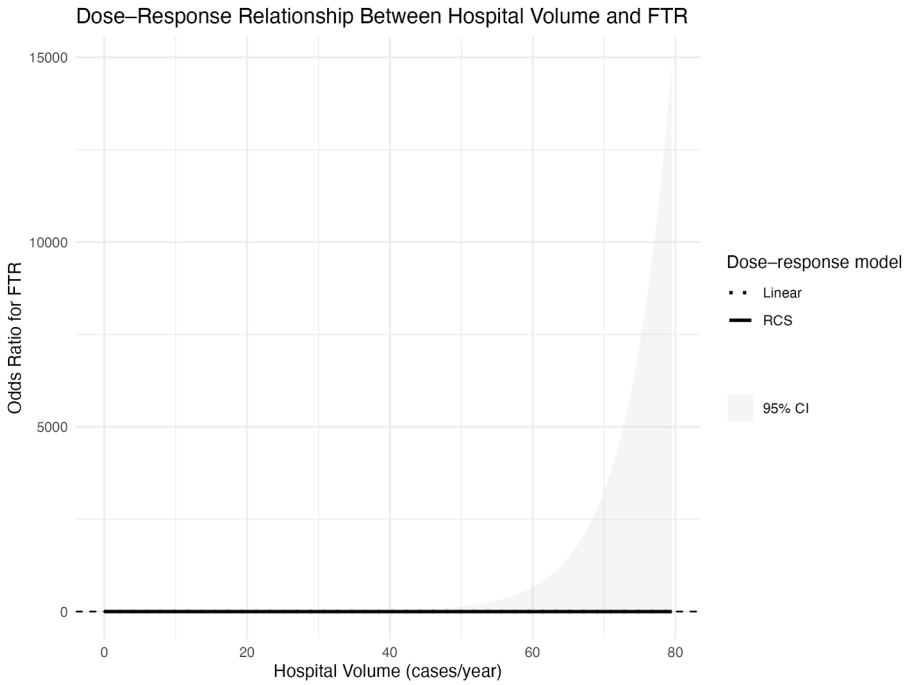


Fig. 3. Dose—response relationship between hospital volume and failure-to-rescue after open repair estimated using restricted cubic splines. The *dotted line* represents the linear trend and the *shaded area* the 95% CI. CI, confidence interval.

Figs. 20 and 21). Similar stability was observed for the remaining analyses, including surgeon volume for open repair mortality, hospital volume for FTR after open repair, and hospital volume for EVAR mortality, with pooled estimates remaining consistent across iterations (Supplementary Figs. 22—24).

Risk of Bias Assessment

As detailed in Supplementary Figure 25, most studies were judged to have high or moderate risk of bias across individual ROBINS-I domains. Bias due to confounding (D1) was the most frequent source of high risk, although it was identified in only a minority of studies. The remaining domains—particularly classification of interventions (D2) and selection of the reported result (D6)—were predominantly rated as low risk. Bias related to missing data (D4) and outcome measurement (D5) was generally assessed as low or unclear. Overall, 3 studies were judged to be at high risk of bias, mainly driven by concerns in confounding or outcome measurement.

DISCUSSION

In this systematic review and meta-analysis including 11 studies, we evaluated the association between procedural volume, mortality and failure to rescue following elective AAA repair. Our main findings were (1) higher hospital-level volume was consistently associated with lower mortality for overall elective repair and failure to rescue for open elective repair; (2) hospitals at the 90th percentile of volume had 35% lower odds of overall mortality than those at the 10th percentile. AAA repair includes cases with varying anatomical complexity, from standard infrarenal to more complex juxtarenal and pararenal aneurysms.[28,29] Because many observational studies analyze these entities together, case mix is often heterogeneous. High-volume centers are also more likely to treat more complex and higher-risk patients, which may partly explain both the persistent hospital-volume effect and the variability across studies.[28]

The observed volume—outcome relationship is consistent with current guideline recommendations supporting centralization of elective AAA repair in experienced centers. However, definitions of ‘‘high volume’’ remain heterogeneous, and the threshold at which outcomes plateau is still unclear. In our dose—response meta-analysis of 6 studies, higher hospital volume was significantly associated with lower overall mortality in a predominantly linear pattern, with no evidence of nonlinearity. Hospitals at the 90th percentile of volume had 35% lower

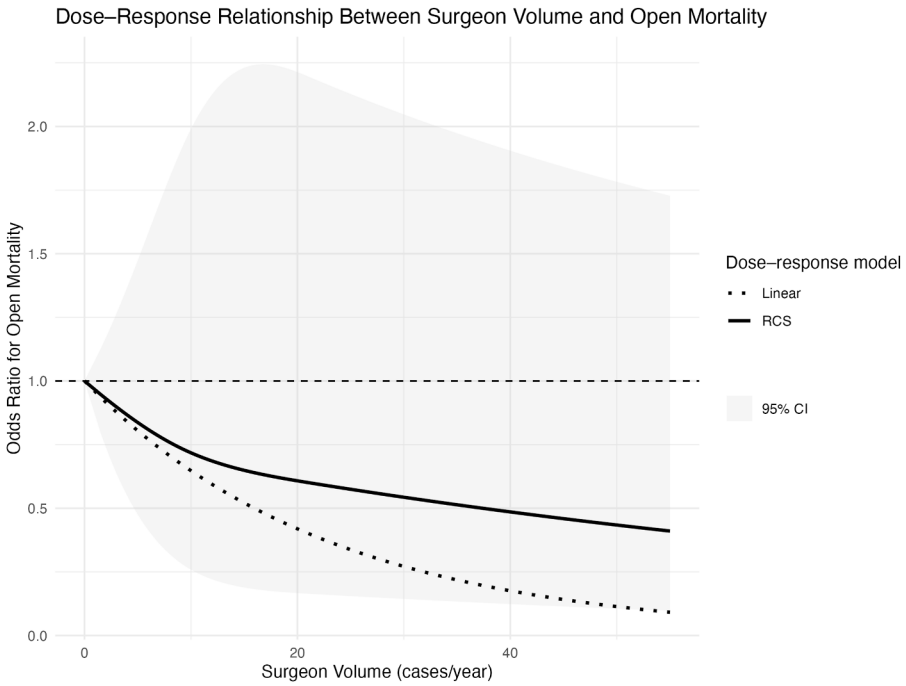


Fig. 4. Dose–response relationship between surgeon volume and open repair mortality estimated using restricted cubic splines. The *dotted line* represents the linear trend and the *shaded area* the 95% CI. CI, confidence interval.

odds of mortality than those at the 10th percentile, and the lowest predicted mortality was observed at approximately 59 procedures per year, although heterogeneity remained substantial. Higher hospital volume was also associated with lower FTR after open repair, suggesting progressively lower odds of death after complications as volume increased. Because the predicted minimum occurred near the upper limit of the observed range, these findings support a largely monotonic relationship rather than a clearly defined threshold, and may reflect greater surgical experience, more structured perioperative care, and better institutional infrastructure.

The 2018 Society for Vascular Surgery guidelines recommend that open AAA repair be performed in hospitals with mortality rates below 5% and at least 10 open repairs annually, while EVAR should be undertaken in institutions with mortality and conversion rates $\leq$2% and at least 10 cases per year.[30,31] More recently, the 2024 ESVS guidelines proposed a minimum volume of 13–16 open repairs annually.[32] Our results suggest that outcome improvements may extend beyond these thresholds, supporting the concept that the volume–outcome relationship is graded rather than threshold-based.[1,33] Nevertheless, centralization strategies must be balanced against considerations of access and equity, as implementation of volume standards may increase travel distances for both urban and rural patients. Therefore, regionalization policies should account for local infrastructure, referral networks, transport logistics, and the geographic distribution of patients.

Our findings align with international evidence that high-volume centers achieve better outcomes. Analysis from the International Consortium of Vascular Registries involving more than 178,000 AAA repairs found that centers performing 13–16 open repairs per year had mortality around 3.1% compared with 4.6% in lower-volume centers, and only 23% of centers met this threshold.[32] The Vascular Quality Initiative registry similarly reported that hospitals performing more than 10 open repairs annually had lower pulmonary and renal complication rates, lower 30-day and 1-year mortality, and about a 50% reduction in FTR compared with low-volume centers.[34,35] These observations suggest that the benefits of higher procedural volume likely reflect not only technical experience but also the organizational and rescue capabilities of high-volume centers.[35]

Hospital volume was associated with better outcomes, particularly lower overall mortality and lower FTR after open repair. Although these

associations were not statistically significant when EVAR and open repair were analyzed separately, the direction of effect remained consistent. This pattern suggests that outcomes may depend more on institutional and structural factors than on the specific repair type. High-volume hospitals may benefit from more organized perioperative pathways, specialized teams, advanced imaging, and greater rescue capacity.[36] Together, these findings support the view that systems of care and institutional resources may be more important than surgeon volume alone, reinforcing the centralization of complex aortic procedures in experienced centers.[37]

Surgeon volume was not significantly associated with mortality after open AAA repair.[1,2,38] Although comparing the 90th and 10th percentiles hinted at a protective effect, wide CIs preclude firm conclusions. Earlier analyses suggested that surgeons performing more than 7 repairs annually achieve lower complication rates and better short-term survival, but evidence on individual surgeon volume remains inconsistent.[2,34,38,39] Our meta-analysis also found no significant association between surgeon volume and mortality or FTR, aligning with studies that highlight institutional factors such as imaging quality, structured perioperative protocols, multidisciplinary teams, and intensive care access as more decisive than individual experience.[40–43] These findings suggest that investing in structured care systems and team training may yield greater improvements than imposing minimum case-volume requirements for individual surgeons.

Temporal meta-regression suggested that the volume–outcome relationship may have attenuated over time. Although study year was not significantly associated with the dose–response slope for overall or open repair mortality, a modest positive trend was observed. In contrast, study year significantly modified the association for EVAR mortality, suggesting a progressive reduction in the volume effect over time. This may reflect greater standardization of EVAR, advances in device technology, and wider dissemination of technical expertise.[44] However, the substantial residual heterogeneity in the overall and open repair analyses indicates that other factors, such as case mix, referral patterns, and institutional resources, likely continue to influence outcomes.

For FTR, mixed-effects meta-regression showed no association between study mid-year and the dose–response slope, suggesting a stable effect over time. Study-level clinical characteristics also did not significantly modify the association or explain the heterogeneity. These findings support a robust inverse association between hospital volume and FTR after open repair, possibly reflecting greater clinical experience, more structured perioperative protocols, more readily available intensive care support, stronger multidisciplinary expertise, and more systematic postoperative monitoring in higher-volume centers.[45] However, the small number of included studies may have limited the power to detect potential effect modifiers.

Sensitivity analyses supported the overall robustness of the findings, but the surgeon-volume analysis for open repair was sensitive to exclusion of the study by Gray et al., which shifted the pooled estimate from significant to nonsignificant.[46,47] This may reflect the distinctive methodological and healthcare-system features of that study, including a highly centralized setting, strict volume thresholds, and extensive adjustment for frailty and institutional clustering. In addition, the smaller number of studies in the surgeon-volume analysis likely increased susceptibility to single-study influence.

Our findings apply specifically to elective aneurysm repair, which differs substantially from ruptured aneurysm repair, where time-critical logistics and immediate rescue capacity play a central role.[28,48] Still, a prior meta-analysis of ruptured AAA repair found a similar volume–outcome relationship, with lower perioperative mortality in high-volume hospitals, particularly for open repair, while surgeon volume was not significantly associated with outcomes.[47] Although these are distinct clinical settings, the consistency of findings supports the importance of hospital-level experience, infrastructure, and multidisciplinary organization in aortic care. These associations should be interpreted cautiously, as they likely reflect not only volume but also case selection, institutional resources, and rescue capacity.

A key methodological challenge was the heterogeneity in how procedural volume was defined across studies, with thresholds based on either absolute annual caseloads or dataset-specific quantiles. To address this, we performed a dose–response meta-analysis using the Greenland and Longnecker method, which allowed us to model the association between volume and outcomes continuously across the full range of reported categories. This approach reduced reliance on arbitrary cutoffs and provided a more informative assessment of whether the volume–outcome relationship follows a graded pattern rather than a simple high-versus-low comparison.

Several limitations should be acknowledged. All included studies were observational and carried moderate to serious risk of bias, mainly due to residual confounding and incomplete adjustment for baseline differences. Definitions of procedural volume varied substantially across studies, and although we addressed this with a Greenland–Longnecker dose–response meta-analysis, this heterogeneity should still be considered. Mortality definitions also differed, with studies reporting either in-hospital or 30-day mortality, which may have contributed to measurement variability. In addition, the predominance of large administrative databases raises the possibility of overlapping cohorts and disproportionate representation of specific healthcare systems.

CONCLUSION

Higher hospital volume was associated with lower overall mortality and lower FTR after open repair, with predominantly linear dose–response relationships. However, no significant associations were found in analyses stratified by repair type or by surgeon volume. Overall, these findings support a predominantly hospital-level volume–outcome relationship and suggest that the benefits associated with higher procedural volume may reflect broader institutional capabilities rather than volume alone, reinforcing the importance of experienced centers, structured perioperative care, and multidisciplinary management in elective AAA repair.

CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

Mayara Leite Coutinho: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lucas Monteiro Delgado:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Mario D'Oria:** Writing – review & editing, Writing – original draft, Supervision, Investigation, Formal analysis, Data curation, Conceptualization. **Salvatore T. Scali:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Conceptualization. **Lydia Hanna:** Writing – review & editing, Writing – original draft, Supervision. **Antonio Eduardo Zerati:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Conceptualization. **Nelson de Luccia:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Conceptualization. **Pedro Puech Leão:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

SUPPLEMENTARY DATA

Supplementary data related to this article can be found at https://doi.org/10.1016/j.avsg.2026.06.025.

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